

1.

$$\lim_{R \rightarrow \infty} \int_0^{\frac{\pi}{2}} e^{-R \sin \varphi} d\varphi = ?$$

$$e^{-R \sin \varphi} \leq e^{-R \frac{2\varphi}{\pi}} \rightarrow 0 \leq \int_0^{\frac{\pi}{2}} e^{-R \sin \varphi} d\varphi \leq \int_0^{\frac{\pi}{2}} e^{-R \frac{2\varphi}{\pi}} d\varphi = - \frac{e^{-R \frac{2\varphi}{\pi}}}{\frac{2R}{\pi}} \Big|_0^{\frac{\pi}{2}} = - \frac{e^{-R}}{2R} \pi + \frac{1}{2R} \pi =$$

$$= \frac{\pi}{2R} (1 - e^{-R})$$

$$0 \leq \int_0^{\frac{\pi}{2}} e^{-R \sin \varphi} d\varphi \leq \frac{\pi}{2R} (1 - e^{-R}) \quad \Big| \cdot \lim_{R \rightarrow \infty}$$

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0

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0

$$\Rightarrow \lim_{R \rightarrow \infty} \int_0^{\frac{\pi}{2}} e^{-R \sin \varphi} d\varphi = 0 \quad \checkmark$$

$$(2) F(y) = \int_y^{\sqrt{y}} \frac{e^{-x^2}}{x} dx \quad y > 0$$

$$F'(y) = ?$$

$$F(y) = \int_y^{\sqrt{y}} \frac{\partial}{\partial y} \left(\frac{e^{-x^2}}{x} \right) dx + \frac{e^{-1}}{\sqrt{y}} \cdot \frac{1}{2} y^{-1/2} - \frac{e^{-y}}{y} =$$

$$= \int_y^{\sqrt{y}} \frac{x e^{-x^2}}{y^2} dx + \frac{1}{2ey} - \frac{e^{-y}}{y} =$$

$$= \int_y^1 \frac{e^{-t}}{2y} dt + -11- = - \frac{e^{-t}}{2y} \Big|_y^1 + -11- =$$

$$= \frac{e^{-1}}{2y} + \frac{e^{-y}}{2y} - \frac{1}{2ey} - \frac{e^{-y}}{y} = \underline{\underline{- \frac{e^{-y}}{2y}}}$$

$$t = \frac{x^2}{y}$$

$$dt = \frac{2x}{y} dx$$

③ $F(y) = \int_0^{\infty} \frac{y dx}{1+x^2 y^2}$

a) za katere $y \in \mathbb{R}$ konv. $F(y)$

b) $F(y) = ?$

c) ali $F(y)$ konv. enakomerno?

d) ali $F(y)$ konv. enakomerno na $[c, d] \subset (0, \infty)$?

e) ali $F(y)$ konv. enakomerno na $(0, d) \subset (0, \infty)$?

a) $F(0) = 0 \rightarrow y=0$ konv.

$$F(-y) = - \int_0^{\infty} \frac{y dx}{1+x^2 y^2} = -F(y) \Rightarrow F(y) \text{ liha}$$

$\rightarrow$ za katere $y > 0$ $F(y)$ konv.?

$\cdot f(x, y) = \frac{y}{1+x^2 y^2}$ kima pola pri $x=0 \rightarrow$ edini problem s konv. je $x \rightarrow \infty$

$$\int_1^{\infty} \frac{y dx}{1+x^2 y^2} \cdot \frac{x^2}{x^2} = \int_1^{\infty} \frac{\frac{y x^2}{1+x^2 y^2}}{x^2} dx$$

$\leftarrow$ ali je omejena funkc.?

$$\lim_{x \rightarrow \infty} \frac{y x^2}{1+x^2 y^2} = \lim_{x \rightarrow \infty} \frac{y}{\frac{1}{x^2} + y^2} = \frac{1}{y} \checkmark$$

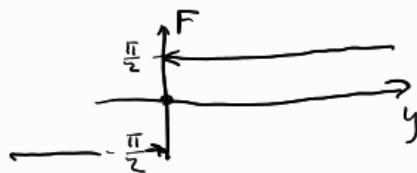
$> 1 \Rightarrow$ integral konvergira

ker konv. na $y \in [0, \infty)$ in F liha $\Rightarrow F$ konvergira za $y \in \mathbb{R}$

b) $F(y) = \int_0^{\infty} \frac{y dx}{1+x^2 y^2} = y \int_0^{\infty} \frac{1}{1+x^2 y^2} dx = y \cdot \frac{\arctg xy}{y} \Big|_0^{\infty} = \frac{\pi}{2}$

$y > 0$

$$\rightarrow F(y) = \begin{cases} \frac{\pi}{2}; & y > 0 \\ 0; & y = 0 \\ -\frac{\pi}{2}; & y < 0 \end{cases}$$



c) Ne, ker $F(y)$ ni zvezna v $y=0$

d) Uporabimo Weierstrassov krit.: $y \in [c, d] \subset (0, \infty)$

$$\left| \frac{y}{1+x^2 y^2} \right| = \frac{y}{1+x^2 y^2} \leq \frac{d}{1+x^2 c^2} \rightarrow \text{ali konv.}$$

$$\int_0^{\infty} \frac{d}{1+c^2 x^2} dx = \frac{d}{c} \arctg cx \Big|_0^{\infty} = \frac{d}{c} \frac{\pi}{2} \checkmark$$

$$\Rightarrow \int_0^{\infty} \frac{y dx}{1+x^2 y^2} \text{ konv. en. na } [c, d]$$

e) Weierstrassov krit.: $y \in (0, d)$

$$\frac{y}{1+x^2 y^2} \leq \frac{d}{1+0} = d \rightarrow \int_0^\infty d \cdot dx = \infty \rightarrow \text{divergira} \rightarrow \text{Weierstr. krit. nam ne da odgovora}$$

→ preverimo konv. na direkten način

$$\left| \int_M^\infty f(x, y) dx \right| < \varepsilon > 0$$

$$\int_M^\infty f(x, y) dx = \arctg xy \Big|_M^\infty = \frac{\pi}{2} - \arctg My$$

ali se da izbrati dovolj velik M , da $< \varepsilon$ za $\forall y \in (0, d)$?

→ Najbo M fiksno poljubno veliko št.

$$\lim_{y \downarrow 0} \left(\frac{\pi}{2} - \arctg My \right) = \frac{\pi}{2} \rightarrow \text{za } \forall M \text{ lahko izberemo dovolj majhen } y, \text{ da je ostarek velik.} \Rightarrow \underline{F \text{ ne konv. enak. na } (0, d)}.$$

→ Enako se rešuje tudi za primere $\int_a^b f(x, y) dx$, ko je f neomejena pri $x=a$

4.) $F(y) = \int_0^{\pi/2} \ln(y^2 - \sin^2 \varphi) d\varphi$

a) Pokaži, da F konv. za $\forall y \in [1, \infty)$

b) pokaži, da je F zvezna na $[1, \infty)$

a) $F(1) = \int_0^{\pi/2} \ln(\cos^2 \varphi) d\varphi = 2 \int_0^{\pi/2} \ln(\cos \varphi) d\varphi = -2 \int_{\pi/2}^0 \ln(\cos(\frac{\pi}{2} - t)) dt = 2 \int_0^{\pi/2} \ln(\sin t) dt.$

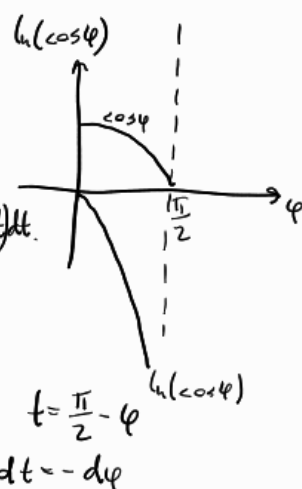
$$2 \int_0^{\pi/2} \frac{\ln(\sin t) dt}{\sqrt{t}}$$

$$\lim_{t \rightarrow 0} \sqrt{t} \ln(\sin t) \stackrel{\text{L'Hopital}}{=} \lim_{t \rightarrow 0} \frac{\ln(\sin t)}{t^{-1/2}} = \lim_{t \rightarrow 0} \frac{1 \cdot \cos t \cdot 2}{\sin t \cdot t^{3/2}} =$$

$$= \lim_{t \rightarrow 0} 2 \cos t \cdot \lim_{t \rightarrow 0} \frac{t \cdot \sqrt{t}}{\sin t} = 0$$

f omejena in $p = \frac{1}{2} < 1 \Rightarrow F(1)$ konv. ✓

če $y > 1$ je $\ln(y^2 - \sin^2 \varphi)$ zvezna, omejena $\Rightarrow F$ obstaja ✓



$$\int_a^b \frac{f(x)}{x^p}$$

- 1) f omejena, $p < 1 \Rightarrow$ konv
- 2) f omejena stran od 0, $p \geq 1 \Rightarrow$ div

b) pogoji: - f zvezna ✓

- F konv. enak. na $[1, \infty)$

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1) ali F konv. enak. na $[1, 2]$?

2) ali F konv. enak. na $[c, d] \subset [2, \infty)$?

2) $\int_0^{\frac{\pi}{2}} \ln(y^2 - \sin^2 \varphi) d\varphi$

$$\underbrace{|\ln(y^2 - \sin^2 \varphi)|}_{\geq 3} \leq \ln(d^2 - \sin^2 \varphi)$$

$$\int_0^{\frac{\pi}{2}} \ln(d^2 - \sin^2 \varphi) d\varphi \quad \text{konv. enak. ✓}$$

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